Finite Groups and Their Representations — Exercise Sheet 7

The group algebra and finite Fourier analysis

Variant: Questions only

Exercise 1 (The trivial and sign idempotents). In $\mathbb{C}[S_3]$ compute $e_{\mathbf{1}}$ and e_{sgn} explicitly. Verify their idempotence and orthogonality by group-algebra multiplication. Find their ranks on the left regular module and on the permutation module $\mathbb{C}^3$. Compare their sum with the identity and with Lecture 7's e_E.

Exercise 2 (Orthogonality of matrix coefficients). Let G be finite. Choose an invariant inner product on each $U \in \mathrm{Irr}(G)$ and an orthonormal basis. Write $D_U(g)_{ij}$ for its matrix entries, called its *matrix coefficients*. Prove

$$\frac{1}{|G|}\sum_{g\in G}\overline{D_U(g)_{ij}}D_V(g)_{kl} = \begin{cases}\delta_{ik}\delta_{jl}/\dim U, & U = V,\\ 0, & U \neq V.\end{cases}$$

Deduce that all functions $\sqrt{\dim U}\, D_U(\cdot)_{ij}$ together form an orthonormal basis of the space of functions $G \to \mathbb{C}$, with the normalised inner product of Lecture 6. Using Lecture 7's basis of class functions by irreducible characters, prove the converse promised on Sheet 6: if every irreducible character is real-valued, then every $g \in G$ is conjugate to g^{-1}.

Exercise 3 (Fourier inversion and convolution). With the unitary matrices D_U in the orthonormal bases of Exercise 2, for $f\colon G \to \mathbb{C}$ define

$$\widehat{f}(U) = \sum_{g\in G} f(g)D_U(g), \qquad U \in \mathrm{Irr}(G).$$

For functions f, h define $(f * h)(x) = \sum_y f(y)h(y^{-1}x)$. Prove the following formulas, with this convention for the forward transform:

$$\widehat{f * h}(U) = \widehat{f}(U)\widehat{h}(U), \qquad f(x) = \frac{1}{|G|}\sum_U \dim U\ \mathrm{tr}\big(D_U(x^{-1})\widehat{f}(U)\big).$$

Also prove the Plancherel identity

$$\sum_x \overline{f(x)}h(x) = \frac{1}{|G|}\sum_U \dim U\ \mathrm{tr}\big(\widehat{f}(U)^\dagger \widehat{h}(U)\big).$$

Exercise 4 (The discrete Fourier transform). Let $C_m = \langle a\rangle$ and $\zeta = \exp(2\pi\mathrm{i}/m)$, with $m \geq 1$. Prove that the map

$$\sum_{k=0}^{m-1} f_k a^k \longmapsto \left(\sum_{k=0}^{m-1} f_k \zeta^{jk}\right)_{0\leq j<m}$$

is an algebra isomorphism $\mathbb{C}[C_m] \cong \mathbb{C}^m$, where the target has coordinatewise multiplication. For $m = 4$, compute the transform of $f = (1, 2, 0, -1)$ and its convolution with $h = (1, -1, 0, 0)$. Writing $\widehat{f}_j$ for the jth transform entry, verify Parseval's identity

$$\sum_{k=0}^{3}|f_k|^2 = \frac{1}{4}\sum_{j=0}^{3}|\widehat{f}_j|^2,$$

the diagonal case of Exercise 3's Plancherel identity.

Exercise 5 (The commutant on two-element subsets). Let X be the six two-element subsets of $\{1,2,3,4\}$, and let S_4 act on $\mathbb{C}^X$. Define operators on its standard basis by

$$Ae_S = \sum_{S' \in X : |S' \cap S| = 1} e_{S'}, \qquad Be_S = e_{\{1,2,3,4\} \setminus S}.$$

Show that $\mathrm{End}_{S_4}(\mathbb{C}^X)$ has basis $\mathbb{1}, A, B$. Compute A^2, B^2, AB, BA and decide whether the representation is multiplicity-free.

Exercise 6 (A skipped restriction step). Let $G = S_5$ and let $H = S_3$ permute $1,2,3$ and fix $4,5$. In $\mathbb{C}[G]$ put

$$a = (1\,4) + (2\,4) + (3\,4), \qquad b = (1\,5) + (2\,5) + (3\,5).$$

Prove $a, b \in \mathbb{C}[G]^H$ and compute $ab - ba$. Independently confirm that restriction is not multiplicity-free using $W_5 = \{x \in \mathbb{C}^5 : \sum_i x_i = 0\}$: prove that it is irreducible for G, and find a repeated irreducible in its restriction to H.

Exercise 7 $\star$. In the orthonormal bases of Exercise 2 put, for $1 \le i, j \le d_U = \dim U$,

$$e^U_{ij} = \frac{d_U}{|G|} \sum_{g} \overline{D_U(g)_{ij}}\, g \in \mathbb{C}[G].$$

Compute every block $D_V(e^U_{ij})$ and prove the multiplication law

$$e^U_{ij} e^V_{kl} = \begin{cases} \delta_{jk} e^U_{il}, & U = V, \\ 0, & U \neq V. \end{cases}$$

Show that $\mathbb{C}[G]\, e^U_{ii}$ is a left submodule of the regular module isomorphic to U. Show that right multiplication by e^U_{ii} is a projection onto this submodule and commutes with the left regular action. For $d_U > 1$, show that left multiplication by e^U_{ii} has a different image and does not commute with that action. Compare $\sum_i e^U_{ii}$ with e_U.